

Bhukya Roopak Krishna

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Professional Summary

AI/ML Engineer who builds and ships production AI systems end to end, from model development to deployment. Built computer vision proctoring and OCR based verification systems used in production, along with LLM powered agentic pipelines for market intelligence, with hands on experience in **LLM fine tuning, RAG, multi agent systems, and MLOps**. Owns the backend, AI workflows, and infrastructure needed to take these systems from prototype to production.

Education

B.Tech, Materials Science & Metallurgical Engineering

2021 – 2025

Indian Institute of Technology (IIT) Hyderabad

Technical Skills

Languages: Python, C, C++, JavaScript, TypeScript

AI / ML: PyTorch, Scikit-Learn, TensorFlow, PEFT, LoRA, NLP, Computer Vision, LLM Fine Tuning

LLM / GenAI: RAG, Agentic AI, Multi-Agent Systems, LangChain, LangGraph, Qdrant, FAISS

MLOps & Cloud: Kubeflow, Kubeflow Pipelines, MLflow, Docker, GitHub Actions, AWS

Backend / Web: React, Next.js, FastAPI, Flask, REST APIs

Databases & Tools: PostgreSQL, MySQL, Git, GitHub, Jupyter, Postman

Professional Experience

AI Full Stack Developer

Nov 2025 – July 2026

Net Connect Global Pvt. Ltd., Bengaluru

- Built an **AI interview proctoring system** that detects suspicious face and voice presence, gaze direction, and mobile devices through computer vision and AI Agents, while also flagging keystroke patterns and unauthorized browser activity during interviews, automatically generating **30+ proctoring reports daily** for review.
- Designed a **KYC verification pipeline** that extracts information from Aadhaar and PAN documents using OCR, with **Mistral AI** handling unclear or incomplete fields, and matches document photos against candidate faces, reducing manual identity checks across **1,000+** production verifications.

Machine Learning Intern

Jun 2025 – Sep 2025

Mobius by Gaian, Hyderabad

- Improved **knowledge graph link prediction by 33%** by replacing R-GCN with CompGCN models and deploying a reusable end to end **MLOps workflow** on Kubeflow Pipelines covering GNN training, evaluation, and versioning.
- Fine tuned **Mistral** and other LLMs on proprietary domain specific datasets using **PEFT** techniques, achieving a **1.4% accuracy gain** in task performance while reducing inference latency by **0.3 seconds**.

Projects

Market Intelligence RAG System | GitHub

Jul 2026 – Sep 2026

- Engineered a **multi agent RAG system** with LangGraph orchestration, LangChain retrieval/tooling, and Qdrant vector search, using Grok for sentiment, event, and market-impact analysis across NIFTY 50 companies.
- Architected **9 specialized agents** and **21 tools** to automate news ingestion, document retrieval, event extraction, and contextual synthesis, transforming real time market developments into structured daily intelligence.

Certificate Verification Platform (SkillKendra) | GitHub

May 2026 – Jul 2026

- Built a **certificate verification pipeline** with FastAPI that extracts text from uploaded certificates using PaddleOCR, EasyOCR, and Tesseract in parallel, then uses **Selenium** to cross check certificate details against official provider records across **50+ certification providers**.
- Added a **Mistral based AI layer** to normalize inconsistent certificate fields and flag potential fraud, with manual review for cases where the OCR output was unclear or incomplete.

Options Trading System – Arbitrage Detection | GitHub

Jan 2026 – Apr 2026

- Built an **ML based options trading system** using **yfinance** to model volatility across different strikes and expiries, achieving **71% directional accuracy** on historical market data, with **MLflow** for experiment tracking.
- Developed a system to spot **pricing discrepancies across options chains**, reaching an **0.84 F1 score** for arbitrage opportunity detection and connecting the results to an end to end data processing and visualization pipeline.

Achievements & Activities

- Contributed to **2 public open source repositories** focused on **machine learning and AI**, with code fixes and improvements.
- Solved **300+ DSA problems** across LeetCode, GeeksforGeeks, and Codeforces.